

Karnataka Board

Biology

Class 12

Instructions:

- i) The question paper consists of four parts – A, B, C, and D

Part – D consists of two sections – Section I and Section-ii

- ii) All the Parts are compulsory.
- iii) Draw diagrams wherever necessary. Unlabeled diagrams or illustrations do not attract any marks.

PART - 1

Answer the following question in one word or one sentence. [10 × 1=10]

1) Name the molecular scissors used in Recombinant DNA technology

Ans: Restriction enzymes are also called 'molecular scissors' as they cleave DNA at or near specific recognition sequences known as restriction sites. These enzymes make one incision on each of the two strands of DNA and are also called restriction endonucleases.

2) What are somaclones?

Ans: The plants produced, that are genetically identical to the original (parent) plant are known as Soma clones.

3) Define reforestation.

Ans: Reforestation refers to the replanting of plants on those land which has been destroyed for the benefits of mankind.

4) Name the enzyme which HIV genome replicates itself in host cell.

Ans: Transcriptase is the enzyme by which HIV genome replicates in the host cell.

5) What is Menarche?

Ans: Menarche is the medical term for a woman's first menstruation, commonly known as her first period. Menstruation is the monthly shedding of the uterine wall. Menarche is a signal that a woman's body is able to become pregnant.

6) **Name the disease diagnosed by widal test.**

Ans: Typhoid is the disease diagnosed by widal test.

7) **Ovulation takes place on the 14th day of menstrual cycle. Why?**

Ans: Ovulation usually takes place 14 days after the beginning of your menstrual cycle. Menstruation occurs after ovulation to clear the tubes.

8) **What is colostrum?**

Ans: Colostrum is a breast fluid produced by humans, cows, and other mammals before breast milk is released. It is very nutritious and contains high levels of antibodies, which are proteins that fight infections and bacteria.

9) **How do some species of insects and frog avoid being detected easily by the Predator?**

Ans: Some species of insects and frogs are cryptically-colored called camouflaged to avoid being detected easily by the predator.

10) **Mention the use of statin.**

Ans: Statins are drugs that can lower your cholesterol. They work by blocking a substance your body needs to make cholesterol. Statins may also help the body reabsorb cholesterol that has built up in plaques on the artery walls thus preventing further blockage in blood vessels and heart attacks.

PART - B

Answer any five of the following questions in 3 – 5 sentences each wherever applicable.

11) **What is inbreeding depression? How is it controlled?**

Ans: Inbreeding is a breeding between individuals who are genetically closely linked such as first and second cousins. Inbreeding leads to the same set of genes or gene pool as it is called to be shuffled between the generations. When inbreeding occurs, recessive genes have a higher chance of being expressed. Inbreeding depression is the decline in the biological physical fitness of the population. This is due to the increased expression of harmful recessive genes.

12) Define biodiversity .Write any two types of biodiversity.

Ans: The existence of a wide variety of plant and animal species in their natural environments, which is the aim of conservationists concerned about the indiscriminate destruction of rainforests and other habitats is called biodiversity.

Two types of Biodiversity are following:

- Genetic biodiversity- Genetic diversity refers to the variation of genes within species. This constitutes distinct population of the same species.
- Species diversity- Species diversity refers to the variety of species within a region it refers to the species richness in terms of number of species in a habitat.

13) Why has T. H.? Morgan selected fruit flies for his genetic experiments?

Ans: T.H. Morgan has selects fruit flies for his genetic experiments due to following reasons: It is easily available hovering over ripe Mango/Banana fruits where it feeds over yeast cells present over the fruit surface.

- The flies can be reared inside bottles having yeast culture over medium containing cream of wheat, molasses and agar
- A new generation can be raised within 2 weeks with single mating producing hundreds of individuals.

14) Differentiate seasonal breeders from continuous breeders.

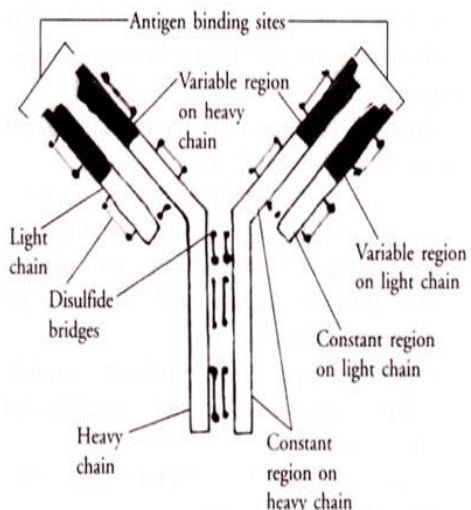
Ans: In continuous breeders like humans the cycle which occurs in the female group is menstrual cycle .Menstrual cycle is a continuous cycle and goes on throughout. Thus those species are called continuous breeders as they can breed at any time of the year. Whereas in seasonal breeders like dogs the cycle which occurs in their female group is oestrous cycle which occurs only in a particular season or particular part of the year. Hence, such organisms are called seasonal breeders as they could breed only in that part of the year when the oestrous cycle takes place in them.

15) What are analogous organs? Give an example.

Ans: Organs similar in shape and function but their origin, basic plan and development are dissimilar. Example. Wings of butterfly bird and bat. Such similarities are because of convergent evolution for adaptation to a common condition.

16) Draw a neat labelled diagram of an antibody molecule.

Ans:



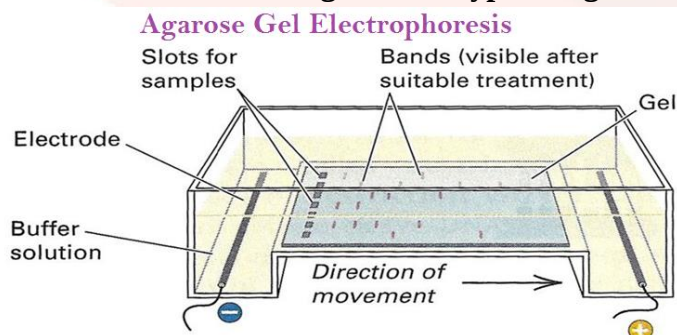
ANTIBODY MOLECULE

17) Name any four recent extinct organism as per IUCN Red List.

Ans: Four recent extinct organism as per IUCN Red List are followings:

1. Madeiran large white
2. West african black rhinoceros
3. Spix's macaw
4. Po'ouli

18) Draw a neat labeled diagram of a typical Agra Rose gel electrophoresis.



PART-C

Answer any five of the following questions in about 40 to 80 words each wherever applicable.
 $5 \times 3 = 15$

19) Explain in brief the separation and isolation of DNA fragments.

Ans: The cutting of DNA by restriction endonucleases results in the fragment of DNA. The technique which separates DNA fragments based on their sizes is called gel electrophoresis. DNA fragments are negatively charged molecules. They can be separated by forcing them to move towards the anode under an electric field through a medium. The most common medium used is agarose. The DNA fragments separate according to their size through sieving effect provided by the agarose gel. The smaller fragment size, faster it moves. The separated DNA fragment can be visualized only after staining that DNA with a compound known as Ethidium Bromide followed by exposure to UV radiation. The DNA fragments can be seen as bright orange bands. These separated bands are cut out from the other rows and extracted from the gel piece. This is called elution.

20) Mention any three characteristics of Cancer cell.

Ans:

- Cancer cells never differentiate. They continue to divide, cause more damage, and invade new tissue which is a result of other pro-survival properties specialized within cancer cells.
- Cancer cells have a loss of contact inhibition. As a result, cancer cells are characterized by uncontrollable growth, the other cells in layers resulting in a tumor, and also invade new tissue. Cancer cells have an asymmetrically-shaped nucleus that is larger than normal, resulting in the reduced presence of cytoplasm within the cell.
- Cancer cells' nuclei have changes in chromatin, abnormalities to structures called mitotic spindle that assist in cell division and contain various genetic abnormalities such as mutations in gene sequencing.

21) What is ecological succession? Distinguish primary succession from secondary succession.

Ans: An orderly sequence of communities of organisms over a period of time at the same place is known as ecological succession. It is the natural process by which the same locality becomes successively re-colonized by different groups of plant communities.

Different Types of Ecological Succession

- Primary succession is initiated when a new area that has never previously supported an ecological community is colonized by plants and animals. This could be on newly exposed rock surfaces from landslides or lava flows
- Secondary succession occurs when an area that has previously had an ecological community is so disturbed or changed that the original community was destroyed and a new community moves in. This is more common than primary succession and is often the result of natural disasters, such as fires, floods, and winds, as well as human interference, such as logging an.

22) What is parthenogenesis give two examples?

Ans: Parthenogenesis is defined as the development of female gamete into new individual without fertilization. The megaspore mother cell undergoes usual meiotic division to form haploid egg and embryo develops from it without fertilization. The megaspore mother cell undergoes the usual meiotic division to form haploid eggs and embryo develops from egg without fertilization. Parthenogenesis can be classified on the basis of mode of reproduction cytological characteristics. The normal forces on parthenogenesis may be obligate in which embryo develops away from unfertilized egg or facultative when embryo develops from both fertilized and unfertilized egg.

23) State Hardy Weinberg principle. Mention any two factors that affect Hardy Weinberg equilibrium.

Ans: Hardy Weinberg's principle states that allele frequencies in a population are stable and are constant from generation to generation.

Factors that affect Hardy Weinberg equilibrium are:

- Genetic migration- When migrations of a selection of population to another place occur, gene frequencies change in the original as well as the new population. New genes/alleles are added to the new population and these are lost from the population.
- Genetic drift- It deals with the gene frequency of a reproducing small population. In a small population not all the alleles, representatives of that species may be present. Thus the inheritance process is in violation of Hardy-Weinberg law.

24) What is artificial hybridization? By which technique is it achieved

Ans: Artificial hybridization is the process in which only desired pollen grains are used for pollination & fertilization. Artificial hybridization is achieved using the following techniques:

- Emasculation:

The anther is removed from bisexual flowers of female parent plant. Before anthers mature. This prevents self-pollination of pollen grains.

- Bagging:

The emasculated flower is Covered prevent contamination with unwanted pollen. Desired pollen grains are dusted on stigma and flowers are bagged again.

25) **With reference to ecological succession define the following terms:**

Climate community - The last succession in a sere is known as climax. The climax is a final and stable community of succession which is controlled by climate. It is capable of producing itself and may not be replaced by other individual as long as climate remains the same. This climax is also known as climate climax.

Sere-During ecological succession the modified environment is a usually LED suitable for the existing community and as such a better suited community replaces it this new community again modify the environment in such a way to make condition less table to itself the changes always keep on taking place in the community structure organisation the sequence of community is termed as Sere.

26) **Write the karyotype of following syndromes.**

i) Down syndrome –

Ans: Human being with a typical karyotype has cells that contain 46 chromosomes, which are grouped into 23 pairs. The karyotype of a person with Down Syndrome reveals cells with a 47th chromosome, brought about by the extra copy of chromosome 21.

ii) Klinefelter syndrome.

People with Klinefelter syndrome have one more chromosome than the normal male. It's XXY instead of XY.

iii) Turner's syndrome-

Besides the numerical abnormalities in Turner syndrome are structural abnormalities of one of the two X chromosomes, homogeneous or mosaic, all fitting as Turner syndrome cytogenetic varieties.

PART-D**SECTION -1**

iv. Answer any four of the following questions in about 200 to 250 words each wherever applicable.

$$4 \times 5 = 20$$

27) Enumerate the salient features of Human Genome Project.

Ans: The salient features of Human Genome Project are followings:

- i) The human genome contains 3164.7 million nucleotide bases.
- (ii) The average gene consists of 3000 bases, but sizes vary greatly, with the largest known human gene being dystrophin at 2.4 million bases.
- (iii) The total number of genes is estimated at 30,000—much lower than previous estimates of 80,000 to 1, 40,000 genes. Almost all (99.9 per cent) nucleotide bases are exactly the same in all people.
- (iv) The functions are unknown for over 50 per cent of discovered genes.
- (v) Less than 2 per cent of the genome codes for proteins.

28) Explain the process of translation in protein synthesis.

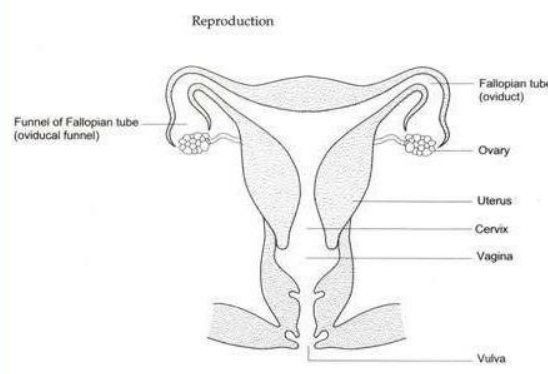
Ans: Protein synthesis is accomplished through a process called translation. After DNA is transcribed into a messenger RNA (mRNA) molecule during transcription, the mRNA must be translated to produce a protein. In translation, mRNA along with transfer RNA (tRNA) and ribosomes work together to produce proteins.

Stages of Translation in Protein Synthesis

- Initiation: Ribosomal subunits bind to mRNA.
- Elongation: The ribosome moves along the mRNA molecule linking amino acids and forming a polypeptide chain.
- Termination: The ribosome reaches a stop codon, which terminates protein synthesis and releases the ribosome.
- Transfer RNA-Transfer RNA plays a huge role in protein synthesis and translation. Its job is to translate the message within the nucleotide sequence of mRNA to a specific amino acid sequence. These sequences are joined together to form a protein.

29) Draw a neat labelled diagram of sectional view of human female reproductive system.

Ans:



FEMALE REPRODUCTIVE SYSTEM

30) a) What are multiple alleles? Give an example. 2

When more than two alternative form of a gene in a population occupies same locus on a chromosome or its homologous are called multiple .Two human examples of multiple-allele genes are the gene of the ABO blood group system, and the human-leukocyte-associated antigen (HLA) genes.

b) Write the symptoms of Klinefelter's syndrome. 3

Ans: Usually, females have two X chromosomes (XX). Males have an X and a Y (XY). But in rare cases, a male is born with an extra X chromosome (XXY). This is Klinefelter's syndrome. It is also called Klinefelter's or XXY. Signs and symptoms of Klinefelter's syndrome vary widely among males with the disorder. Many boys with Klinefelter's syndrome have few noticeable signs, and the condition may go undiagnosed until adulthood. For others, the condition has a noticeable effect on growth or appearance.

31) What is infertility? How is infertility treated by assisted reproductive Technologies like GIFT and IFCI?

Ans: Infertility is the inability of a person, animal or plant to reproduce by natural means. It is usually not the natural state of a healthy adult. There are many treatments and procedures that come under the umbrella classification of assisted reproduction techniques.

GIFT- Gamete intra fallopian Transfer

The sperm and the eggs are combined outside the body and then directly inserted into the fallopian tube via a small incision. The fertilization happens inside the body.

ISCI- Intra cytoplasmic sperm injection.

In this case, the sperm is directly injected into the egg rather than placing the sperm next to the egg. This increases the odds of fertilisation. This procedure is best suited for couples who have decreased sperm count, low motility of the sperm, and high numbers of abnormal sperm.

32) Explain the structure of megasporangium.

Ans: The ovule or mega sporangium has a massesparenchymatous body called nucellus. It is enclosed and protected by one or two sheaths known as integuments. The integument completely cover the nucellus leaving only a small opening called micropile. The ovule is attached to the placenta by a stalk known as funicle. The point of attachment of body of ovule to the funicle is called hilum. The basal part of the ovule where nucellus, integument and funicle merge is known as chalaza.

SECTION –D

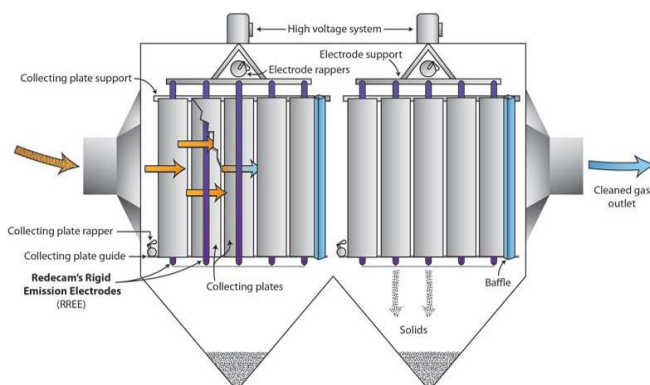
v) Answer any three of the following questions in about 200 to 250 words each wherever applicable.

$$3 \times 5 = 15$$

33) Write a brief account of functioning of electrostatic precipitator with diagram.

Ans: An electrostatic precipitator (ESP) is an device that removes dust particles from a flowing gas (such as air) using the force of an induced electrostatic attraction (i.e, like charges repel; unlike charges attract) Electrostatic precipitators are highly efficient filtration devices that allow the flow of gases through the device, and can easily remove fine particulate matter such as dust and smoke from the air stream.

An ESP uses a high voltage electrostatic field to separate dust, fume or mist from a gas stream. The precipitator consists of vertical parallel plates (collecting plates/electrodes) forming gas passages 12 to 16 in. (30.5 to 40.6 cm) apart. Discharge electrodes are electrically isolated from the plates and suspended in rows between the gas passages. Every particle either has or can be given a charge - positive or negative. A high voltage system provides power to the discharge electrode to generate an electrical field. The particulate, entrained in the gas, is charged while passing through the electrical field. The particulate is then attracted to the grounded collector plate, and forms a dust layer on the plate. Periodic rapping separates the accumulated dust layer from both the collector plates and discharge electrodes (in case of wet ESP by spraying it with a liquid). The dust layer released by the rapping collects in hoppers and is removed by material / ash handling system. At many places in this article, ash is used instead of particulate matter since major application of an ESP is for ash collection.



ELECTROSTATIC PRECIPITATOR

34) a. What is gene therapy? Give an example.

2

Ans: Gene therapy is an experimental technique that uses genes to treat or prevent disease. In the future, this technique may allow doctors to treat a disorder by inserting a gene into a patient's cells instead of using drugs or surgery. Researchers are testing several approaches to gene therapy, including. Gene therapy is currently being tested only for diseases that have no other cures.

b. Write a note on biopiracy with reference to basmati rice.

3

Ans: Bio piracy is the term used to refer to the use of bio resources by multinational companies and other organisations without proper authorisation from the countries and people concerned without compensatory payment and American company got patent rights on basmati rice through the US patent and Trademark office this allowed the complete you celebrate your Basmati in the US and abroad this new variety of Basmati has actually been derived from Indian farmers variety India Basmati was crossed with semi dwarf varieties and claimed as an invention or inability IT results in selling of best basmati rice is restricted.

35) What is inbreeding? Mention the significance of inbreeding.

Ans: Inbreeding is the mating of individuals or organisms that are closely related through common ancestry. Inbreeding is useful in the retention of desirable characteristics or the elimination of undesirable ones but it often results in decreased vigour, size, and fertility of the offspring because of the combined effect of harmful genes that were recessive in both parents. The closest type of inbreeding is selfing, or self-fertilization, the union of male and female sex

cells produced by the same organism the backcross, crossing a first-generation hybrid with one of the parental types is a common method of inbreeding. The adverse effects of inbreeding in animals are well known. The incidence of metabolic disorders, structural abnormalities and inherited disease conditions, caused by harmful recessive genes, increases following inbreeding. Performance in several characters, particularly those concerned with reproduction and survival, declines following the mating of close relatives. This is known as inbreeding depression. These effects are mainly due to an increase in the frequency of homozygous genotypes (AA and aa) at the expense of heterozygotes (Aa), which is caused by inbreeding. It is only harmful, however, when the dominance is directional, which means that the undesirable member of a pair of genes is usually recessive. When a high proportion of these harmful genes are present in the heterozygous state (Aa) the animal is protected from their debilitating effects by the dominance of the normal gene; but when some of the heterozygotes are replaced by homozygous recessives (aa), following inbreeding, their harmful effects become manifest. Other types of gene action are sometimes responsible for inbreeding damage, but are thought to be less important. These are: overdominance, epistatic interaction and the overall level of heterozygosity.

36) Explain the role of microbes in Industrial Products.

Ans: Microorganism possesses several characteristics which make them ideal organism for industrial processes they contain a broad variety of enzymes making an array of chemical conversion possible they have a relatively high metabolic activity which allows conversion to take place rapidly in an industrial process a microbe interaction represents a many chemical factory the efficiency of this factory depends upon Liberation of large amount of single product that may be effectively isolated and purified production on industrial scale requires growing microbes in a very large vessels called fermenter. The commercial products made by microbes include

- Medicines – Microorganism are used in manufacture of several medicines got prepared from a fungus *clavicepspurpurea* is used to induce uterine contraction for abortions menstrual this order and to checked hemorrhages.
- Antibiotics- Penicillin was the first antibiotic to be discovered and it was a chance Discovery .antibiotics have greatly improved our capacity to treat deadly diseases such as played whooping cough leprosy which used to kill millions all over the globe. The first of modern antibiotics is earth Penicillin. Natural penicillin are produced from growth of *Penicilliumnotatum*.
- Vitamins- Microbial fermentation are used for commercial production of several *Ashbyagossypi* and *Eramothecium*.
- Vaccines -mutant strains and at inactivated violent pathogens are used for producing vaccines

- Organic acids – Several organic acid such a Citric, itaconic, gluconic and malic acid are produced on industrial scale true microbial transformation. Gluconic acid is produced by many aspergilli and Penicillia and itaconic acid by Aspergillusitaconicus.

37) a) What is adaptation ?1

Ans: Adaptation is a change or adjustment in structure or habits by which a species becomes better able to function in its environment, occurring through the course of evolution by means of natural selection. Living things are adapted to their habitats. This means that they have special features that help them to survive. An African elephant, for example, lives in a hot habitat and has very large ears that it flaps to keep cool.

b) Write the causes and symptoms of altitude sickness .2

Ans: Altitude sickness is a disorder caused by being at high altitude, where oxygen levels are low, without gradually getting used to the increase in altitude. It is also known as acute mountain sickness (AMS), altitude illness. The condition occurs at altitudes higher than 8,000 feet (ft), or 2,500 meters (m).

The severity of symptoms depends on several factors, including the age, weight, blood pressure, and general fitness of an individual how fast a person climbs to 8,000ft, the amount of time spent at a high altitude. The primary symptom of altitude sickness is a headache.

c) What is brood parasitism .Give an example? 2

Ans: Brood Parasitism is defined as the interaction in which a parasitic bird species places its eggs in the nest of a host species with the intention of freeing itself from parental obligations. The parasitic species only has to find a nest to place its eggs and will not see its offspring again. The parasite is lodged in the host for the purpose of obtaining food, protection, and reproductive opportunities. In the case of brood parasitism in birds, the parasitic species does not obtain food, nor does it depend on the host species for other functions, but only to take charge of incubating the eggs and feeding the chicks of the parasitic bird.